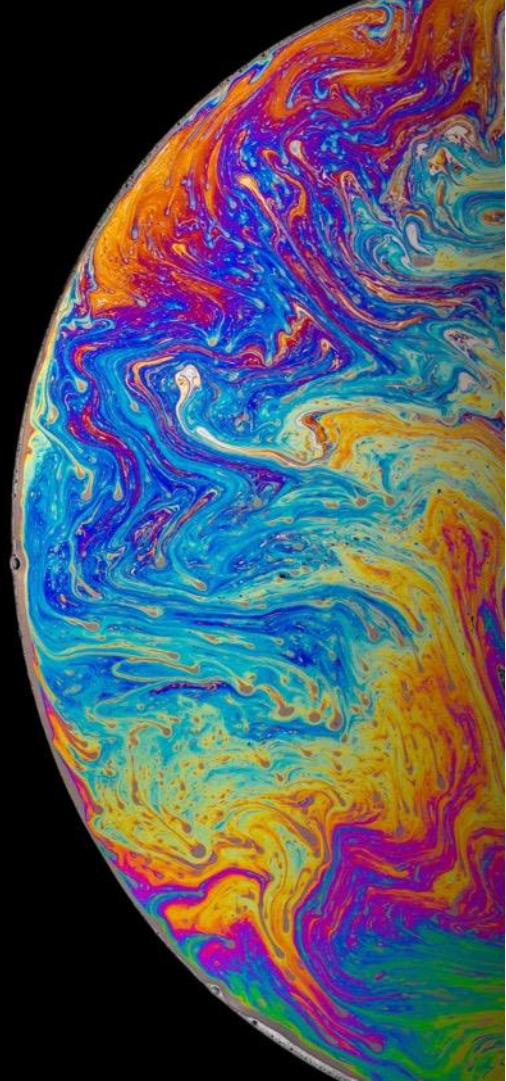
CMIP
Coupled Model Intercomparison Project

CDO and NCO
Working together

Hands-on session

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Installation on macOS or Linux

macOS with Homebrew

```
Homebrew: brew.sh

brew install cdo nco

Ubuntu / Debian
sudo apt update
sudo apt install cdo nco
```

Version checks

cdo -V
ncks --version

Run these in Terminal or iTerm on macOS.

On Linux, open a terminal.

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Installation on Windows

1. PowerShell as administrator

```
wsl --install
```

```
2. Open Ubuntu after restart
sudo apt update
sudo apt install cdo nco
```

Linux commands

Use the Ubuntu terminal for all workshop commands.

cdo -V
ncks --version

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Data and working folder

The same two ACCESS-CM2 files

```
data/historical.nc
data/ssp585.nc

mkdir mixed

cdo showname data/historical.nc
```

Shared definition

1979–2024
Arctic 65–90°N
Baseline 1981–2010

Equal grid-point weights
Equal monthly weights

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1. Join and select the years

NCO: ncrcat

```
ncrcat -v tas \
  -d time,1979-01-01,2024-12-31 \
  data/historical.nc data/ssp585.nc \
  mixed/merged.nc
```

Check the result

```
cdo ntime mixed/merged.nc
```

Expected: 552

One command

Read historical first, then SSP5-8.5.

Keep tas and the selected dates.

552 monthly records

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2. Select the Arctic

CDO: sellonlatbox

```
cdo -b F64 sellonlatbox,0,360,65,90 \
  mixed/merged.nc mixed/arctic.nc
```

Check the result

```
cdo ngridpoints mixed/arctic.nc
```

Expected: 3840

Coordinates

Longitude: 0–360°
Latitude: 65–90°N

-b F64 keeps double precision for the following calculations.

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3. Average the grid points

CDO: fldmean

```
cdo fldmean,weights=false \
  mixed/arctic.nc \
  mixed/arctic_monthly.nc
```

Check the result

```
cdo outputtab,date,value -seltimestep,1/3 \
  mixed/arctic_monthly.nc
```

First month: approximately 242.360 K

Equal weights

Each selected grid point has the same weight.

The output contains one Arctic mean for each month.

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4. Average each year

CDO: yearmean

```
cdo yearmean \
  mixed/arctic_monthly.nc \
  mixed/annual_temperature.nc
```

Check the result

```
cdo ntime mixed/annual_temperature.nc
```

Expected: 46

46 annual values

Each year contains
12 complete months.

Every month has
the same weight.

Temperature stays in K.

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5. Calculate the reference

NCO: ncra

```
ncra -d time,1981-01-01,2010-12-31 \
  mixed/annual_temperature.nc \
  mixed/reference.nc
```

Check the result

```
cdo output mixed/reference.nc
```

Expected: approximately 257.242 K

30 reference years

ncra averages records.
Here, each record
is one year.

The result is one
reference temperature.

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6. Subtract the reference

CDO: sub

```
cdo sub mixed/annual_temperature.nc \
  mixed/reference.nc \
  mixed/annual_anomaly.nc
```

Check the result

```
cdo outputtab,date,value -seltimestep,1/3 \
  mixed/annual_anomaly.nc
```

1979: 0.298596 1980: −0.955143 1981: −0.489441

Temperature difference

Annual temperature minus the reference

A difference of 1 K equals a difference of 1 °C.

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7. Label the anomaly

NCO: ncatted

```
ncatted -a units,tas,o,c,degC \
  -a standard_name,tas,o,c,air_temperature_anomaly \
  -a cell_measures,tas,d,, \
  mixed/annual_anomaly.nc
```

Check the result

```
cdo showunit mixed/annual_anomaly.nc
```

Expected: degC

Metadata

Units: degC
Variable: temperature anomaly

Remove the unused area-measure reference.

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The same annual anomalies

Three values to check

```
1979        0.298596 °C
1980       -0.955143 °C
2024        2.647191 °C

46 years, 1979–2024
```

Check the result

```
cdo output -timmean -selyear,1981/2010 \
  mixed/annual_anomaly.nc
```

Expected: approximately zero

Numerical agreement

Mixed workflow versus independent Python:

Maximum difference about 1×10^{-13} °C

1981–2010 mean ≈ 0

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Your turn: the reference

One small substitution

Replace step 5 with a CDO command.

Select 1981–2010 first.
Then average over time.

Save a second reference file:
mixed/reference_cdo.nc

Discuss the result

Which operations did NCO combine?

Why could we keep temperatures in K?

Which weights must remain unchanged?

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Conda and common problems

Alternative for macOS, Linux or WSL

```
Miniforge: conda-forge.org/download

conda create -n climate -c conda-forge \
  --override-channels cdo nco
conda activate climate

cdo -V
ncks --version
```

If a command fails

Command not found:
activate the environment.

File not found:
check the working folder.

Output already exists:
use a new output folder.

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